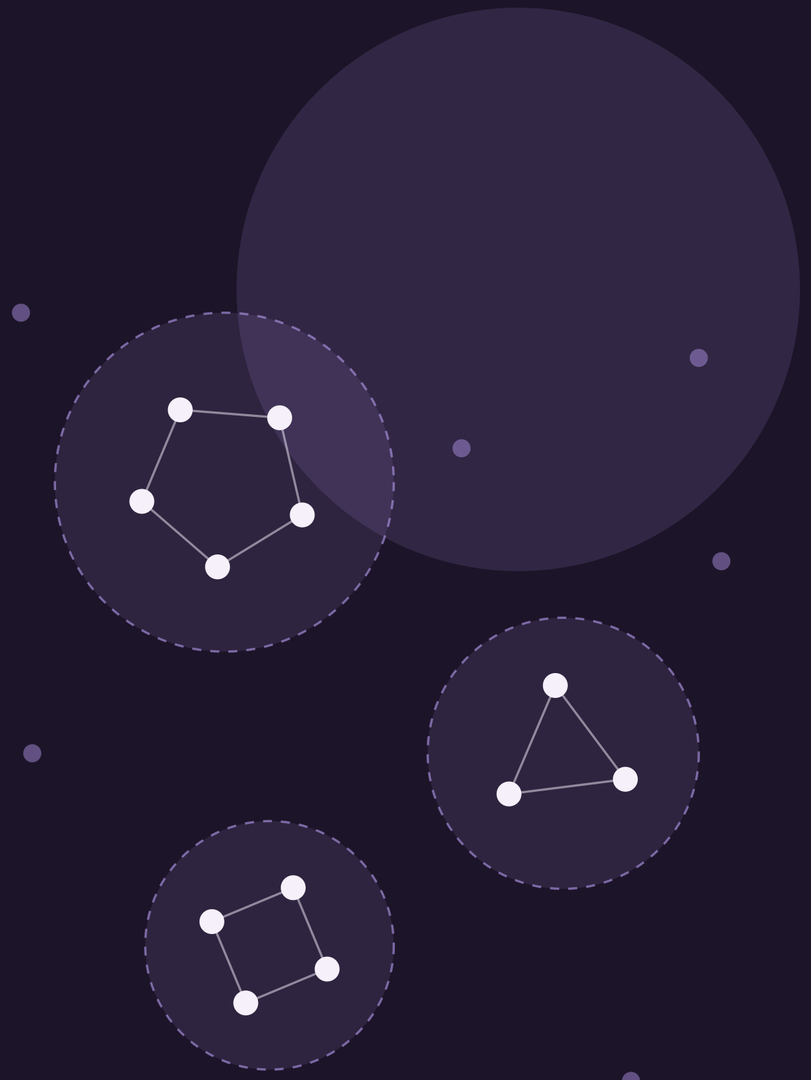
CASE STUDY · NITYA A. KUMAR · OCT 2026

# Same loan, *different rows.*

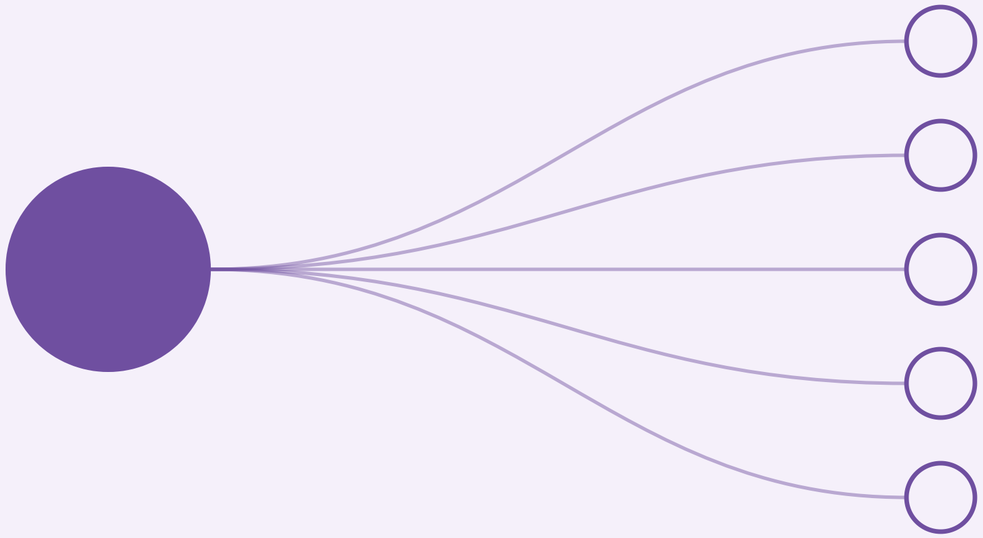
A take-home data case study: re-grouping about 20,000 split mortgage records into real loans, using AI on purpose.



THE PROBLEM

# One loan. *Many rows.*

County records log one row per property, so a single loan can show up several times. My job: figure out which rows are the same loan. No answer key, no rules given.



→ Multi-property loans

→ Loans split into tranches

→ Loans relabeled later

WHAT A RECORD LOOKS LIKE

# Spot the *same loan.*

Made-up example, not the real data. Same lender + same amount + close dates = one loan.

| Row | Deal name | Category | Amount | Date | Lender ID | Result |
|---|---|---|---|---|---|---|
| **A** | MAPLE 1042-RVR | Private credit | $250,000,000 | Mar 3 | 1042 | **Loan 1** |
| **B** | MAPLE 1042-OAK | Private credit | $250,000,000 | Mar 3 | 1042 | **Loan 1 · R1** |
| **C** | MAPLE 1042-PNE | CMBS | $250,000,000 | Mar 6 | 1042 | **Loan 1 · R2, relabeled** |
| **D** | HARBO 2210-RVR | Private credit | $250,000,000 | Mar 3 | 2210 | **Separate: other lender** |
| **E** | MAPLE 1042-ELM | Private credit | $250,000,000 | Apr 20 | 1042 | **Separate: 48 days later** |

THE BRIEF

# AI-assisted *by design.*

The test wasn't whether I could code it. It was whether I could steer AI, check its work, and catch it when it's wrong.

THE SETUP

# 7 agents. *1 skeptic.*

I split the work into AI agents with one job each. Each handed files, not chats, to the next.

1. Profiler
2. Name parser
3. Signal analyst
4. Matcher
5. CMBS linker
6. Validator

7 **Adversarial advocate:** started fresh each time and re-ran every number with its own code.

THE SIGNAL

# Same amount, same day?

For pairs of records with the exact same loan amount: how often were they recorded on the same day?

**52.6%**

when it's the **same lender**

**0.7%**

when it's **different lenders**

Amount alone means nothing. **Lender + amount + date** is the fingerprint.

THE METHOD

# Three rules.

Records that link become one group. Uncertain matches get flagged, never forced.

## R1

**Same lender, same amount, overlapping dates**

The strongest match

## R2

**Same lender, same amount, up to 7 days apart**

Lower confidence

## R3

**No lender ID? Name text + amount, ≤7 days**

With a guard against generic names

WHAT THE SKEPTIC CAUGHT

AI can state a wrong number with total confidence. Before I submitted, agent 7 flagged:

6

wrong or unsupported claims

1

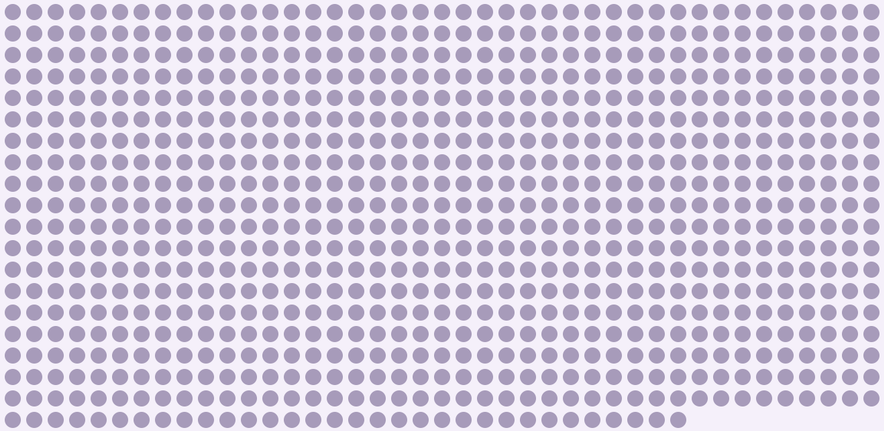
real bug in a rule

The bug: the generic name “FIRST” chained 6 different lenders into one group. Fixed before submitting.

MY CALL

# How common, *not how round.*

A round amount isn't suspicious on its own. What matters is how many records share it.



**$10M** · 811 records · weak clue



**$1.01B** · 7 records · strong clue

SANITY CHECK

One lender, one day, 24 records of $500M each. If those are separate loans, that's...

# $12B?

The busiest lender-day in the whole file was **under half that.** So it's one big portfolio loan, not 24.

CHECKING IT MYSELF

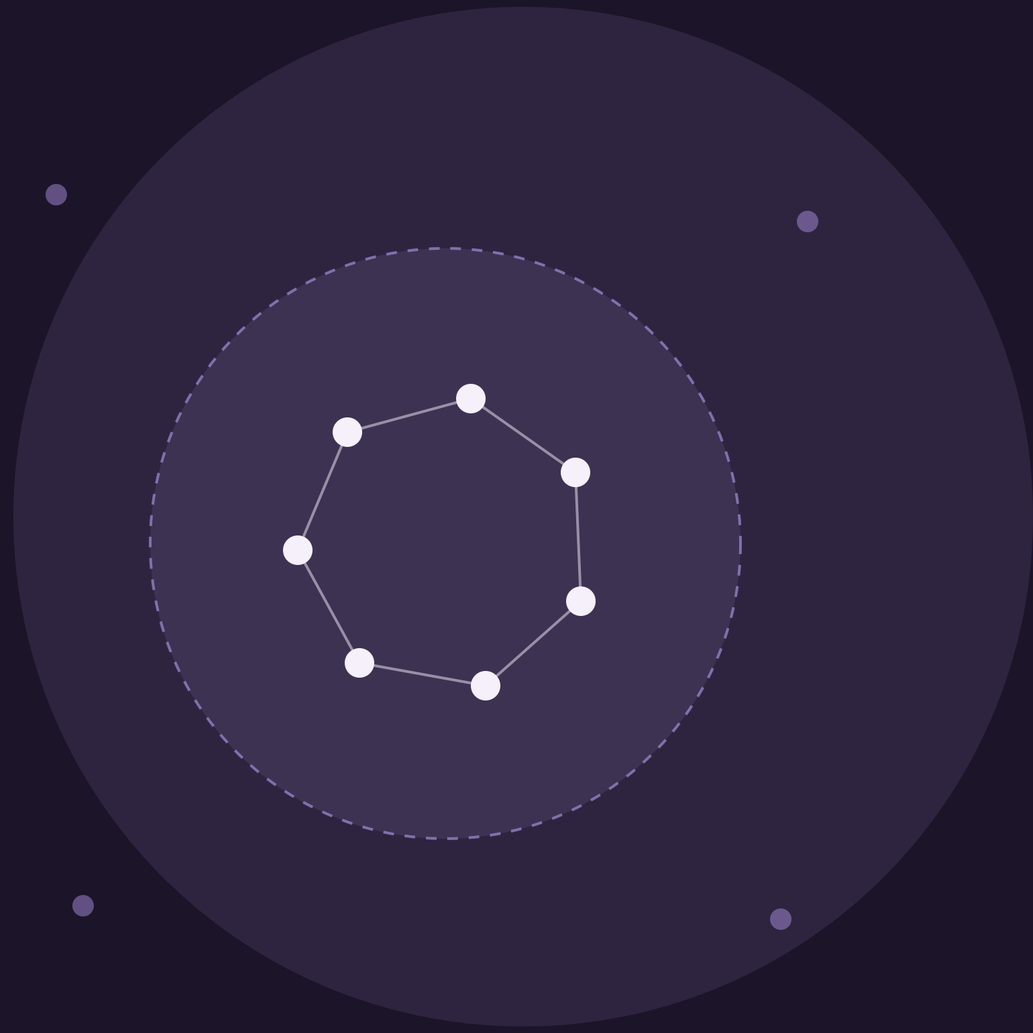
# My verdict *before* the AI's.

I hand-checked 10 risky cases and made my call before looking at what the AI said.

| 6 | 2 | 0 | 1 |
|---|---|---|---|
| merged and correct | merged, unsure | wrongly merged | missed match |

TAKEAWAY

# AI does the volume.
# *I own the judgment.*



~20,000 records → ~17,200 loans. Data and code stay private with the company.